# John Michael Zuniga

jmt.zuniga.z@gmail.com | https://www.linkedin.com/in/john-michael-zuniga-106982335/
https://github.com/water-mizuu | +63 998 546 2358

## Summary

Computer Science graduate (Magna Cum Laude, DOST Scholar) specializing in embedded hardware, ML/AI systems, and full-stack development. Experienced in bridging physical hardware with local network architectures, optimizing neural networks for edge deployments, and delivering production-grade applications across Flutter, Python, and modern web ecosystems.

## Work Experience

**Ateneo Innovation Center** - Software Developer Intern — January 2026 - March 2026

- Developed data transformation and cleaning pipelines in Python, normalizing raw incoming readings into structured formats for consistent storage and downstream consumption.
- Designed and managed a NoSQL schema for time-series environmental data, and handled on-premises deployment of the application stack ensuring reliable local hosting.
- Built a full-stack monitoring interface using React, WebSockets, and REST APIs to stream live sensor telemetry from embedded microcontrollers to an online dashboard, enabling real-time remote environmental visibility.

## Education

**Technological Institute of the Philippines - Quezon City**, BS in Computer Science — August 2022 - April 2026

- DOST Science Education Institute - Merit Scholar
- Magna Cum Laude (GPA of 1.31)

## Projects

**Online ML-Based EEG ADHD Discrimination** | Flask, Bootstrap, Docker, AWS, PostgreSQL — August 2025 - April 2026

Graduate thesis applying Bayesian Optimization to CNN-LSTM hyperparameter tuning for non-invasive ADHD diagnosis in children using sparse EEG recordings.

- Boosted CNN-LSTM model generalizability by 30% through targeted Bayesian Optimization of hyperparameters on sparse EEG recordings.
- Surpassed baseline model performance with above 90% accuracy on the test dataset, demonstrating strong discriminative capability.
- Hosted Python-backed Flask-based web application with real time response rates and continuous user experience without extensive loading screens.

**East Hardware Product Management System** | Flutter, SQLite, WebSocket — October 2024 - July 2025

*A comprehensive, desktop-optimized inventory and sales management system for hardware retail, featuring automated inventory optimization algorithms, custom-built cryptographic security, and a multi-device local network synchronization architecture.*

- Enabled real-time data consistency across 4+ networked devices by developing a scalable local server-client synchronization system using WebSockets and a proxy-based database pattern.
- Delivered a premium, native-feeling desktop experience by architecting the application with Clean Architecture, the BLoC pattern, and the Windows Fluent UI design system.

**Sudoku Solver** | Vue 3, TypeScript, Vite, CI/CD Github Actions August 2024 - December 2024

*An interactive Sudoku solver visualization built with Vue 3 and TypeScript that demonstrates the Wave Function Collapse algorithm enhanced with backtracking and minimum entropy heuristics.*

- Implemented a recursive backtracking mechanism to ensure algorithmic completeness, enabling the solver to exhaustively search and resolve contradictory states.
- Developed a dynamic visualization engine using the Vue 3 Composition API to illustrate constraint propagation, highlighting real-time changes in cell superpositions and backtracking events.
- Architected a custom event-driven state engine to manage asynchronous solver execution, featuring adjustable speeds, step-by-step playback, and detailed action logging.

**Happy Habitat** | Flutter, SQLite, CI/CD Github Actions August 2024 - December 2024

*Happy Habitat is a gamified habit tracking application built with Flutter and SQLite that motivates users to build consistent routines by rewarding them with virtual currency to customize and expand interactive isometric habitats for digital pets.*

- Implemented a custom coordinate transformation algorithm for isometric grid rendering, including real-time depth sorting (Z-ordering) and interactive drag-and-drop placement logic for game assets.
- Architected a persistent local data layer using SQLite, managing complex relational schemas for habit schedules, historical activity logs, and multi-room habitat configurations.
- Engineered a reactive UI with custom-built components, including a performance-optimized horizontal scrolling calendar and a time-partitioned habit categorization system (Morning/Afternoon/Evening).
- Integrated reactive state management using the Provider pattern to synchronize habitat visual states with habit completion progress seamlessly.

**Identipie** | Python, YOLOv8, OpenCV, Tkinter, PIL August 2024 - December 2024

*A real-time food recognition and nutritional analysis application that utilizes a custom-trained YOLOv8 computer vision model to identify food items and provide instant dietary information.*

- Developed a real-time object detection system leveraging the YOLOv8 architecture to identify food categories (e.g., proteins, carbohydrates, fruits) via a live camera feed.
- Engineered a custom training and fine-tuning pipeline using transfer learning, optimizing detection performance for a specialized dataset of nine distinct food classes.
- Integrated a comprehensive nutritional database mapping detected objects to specific macronutrient data, including calories, fats, and proteins, for immediate dietary feedback.
- Designed an interactive desktop GUI using Tkinter and OpenCV, featuring asynchronous frame processing, confidence-based filtering, and real-time bounding box overlays.

**Abandoned** | Java, LibGDX, Gradle June 2024 - December 2024

A procedurally generated top-down adventure game built with LibGDX that utilizes an advanced Wave Function Collapse (WFC) algorithm with recursive backtracking to create coherent and diverse game environments.

- Developed a high-performance procedural generation system using the Wave Function Collapse (WFC) algorithm, enabling the creation of logically consistent game worlds from simple tile adjacency rules.
- Implemented a recursive backtracking state engine to resolve algorithmic contradictions during the collapse process, ensuring 100% convergence on valid map states.
- Engineered a constraint propagation engine using bitwise superposition manipulation to efficiently manage tile possibilities and adjacency constraints across large grids.
- Integrated a spatial actor separation algorithm (similar to Boids) to strategically place key gameplay landmarks and rescue zones prior to the procedural collapse.
- Designed a dynamic mission probability engine using breadth-first search (BFS) to calculate accessibility and "rescue probability" heatmaps across the generated terrain.

## Certifications

**Google IT Automation with Python Professional Certificate** February 2026

- Demonstrated proficiency in Python automation, Git/GitHub version control, and system administration to streamline IT workflows and troubleshoot complex network issues.

**TOPCIT Level 4 (Proficient) 12th & 13th Philippines** June 2025 & December 2025

- Achieved Level 4 (Proficient) competency with a score of 677, placing in the top 5% of all national TOPCIT Philippines participants across both the 12th and 13th administrations.

## Awards & Honors

Consistent honor student: President's Lister (3×), Dean's Lister (1×), VPAA Lister (3×) across all four academic years.

## Skills

**Languages:** Python, Java, TypeScript, JavaScript, Dart, C++, C#, PHP, Rust, Zig

**Frameworks & Libraries:** Flutter, PyTorch, OpenCV, YOLOv8, NumPy, Pandas, ReactJS, NextJS, VueJS, .NET, LibGDX

**Databases:** PostgreSQL, SQLite, Firebase, Supabase

**Cloud & DevOps:** AWS, Azure, Vercel, Docker, CI/CD, Git, GitHub, Linux/Bash

**Hardware & Embedded:** Raspberry Pi, Arduino, ESP32

**Design & Tooling:** Figma, Jupyter Notebook, Jest, Auth.js

**Spoken Languages:** English, Tagalog, Japanese